

2024



Bluecurrent Creates a Flexible Smart Metering Solution Working with VTS and AWS

2

Learn how Bluecurrent met changing smart meter regulations with enhanced data processing using Diverge from AWS Partner VTS.

[Overview](#) | [Opportunity](#) | [Solution](#) | [Outcome](#) | [AWS Services Used](#)

Processes 6x data 10x faster

to comfortably meet regulatory obligations and customer requirements

Facilitates

improved analytics in a secure environment

Improves resilience

and disaster recovery capabilities

Provides a foundation

to facilitate future innovation

Overview

With new energy industry regulations requiring [Bluecurrent](#) to handle six times more meter data, the smart meter services company knew that it was time to upgrade to a scalable cloud solution. Bluecurrent saw an opportunity alongside this change in regulatory requirements to offer faster data processing and better analytics to help its customers keep up with market trends, such as the growing popularity of electric vehicles and renewable energy sources.

Bluecurrent turned to [Vector Technology Solutions](#) (VTS), which formed a strategic alliance with Amazon Web Services (AWS) to co-develop Diverge, a next-generation energy data and analytics solution. Using Diverge, Bluecurrent has not only enhanced its data processing capabilities but also improved its service delivery with faster processing speeds and improved analytics, underpinned by a highly resilient and secure infrastructure.





Opportunity | Using AWS to Develop a Scalable Meter Data Solution for Bluecurrent

Bluecurrent has been providing data services and smart meters to electricity and gas distributors and retailers since 2007. It has a fleet of more than two million meters in Australia and New Zealand. In 2017, the Australian Energy Market Commission passed a rule called the 5-Minute Settlement, requiring smart meters to measure electricity consumption in 5-minute intervals, compared to the previous 30-minute requirements, with the change coming into effect in 2021. Bluecurrent needed to update its systems to comply with the new requirement, and it also wanted to be able to offer customers richer data services beyond billing to help them meet decarbonization goals.

To create the solution it needed, Bluecurrent looked to VTS, an [AWS Partner](#). “We needed a solution that would let us flex as new requirements emerged,” says Anna Carrick, general manager of data services for Bluecurrent. “Within the energy market, there is rapid change coming from distributed generation, electrification, and smarter homes, so we need to have an architecture we can adapt for new use cases moving forward.”

VTS chose to develop a software-as-a-service solution using serverless technology from AWS to provide the scalability and resiliency that Bluecurrent needed. VTS and AWS formed a strategic alliance in 2020 to create a new solution. The result—called Diverge—went live after achieving accreditation against the new 5-Minute Settlement rules. “We passed accreditation with flying colors, in a fraction of the normal time,” says Ray Cheung, subject matter expert in metering data for Bluecurrent. As of January 2024, Bluecurrent is using Diverge for more than one million meters in Australia and New Zealand.

Solution | Processing 6x the Data 10x Faster While Improving Resilience Using Diverge

Diverge is an entirely serverless energy data solution built on new and existing AWS services using infrastructure as code and a microservices architecture. The VTS and AWS teams collaborated along with subject matter experts from Bluecurrent and others in the industry to understand customer and market needs. “We had outstanding collaboration, not only through the strategic alliance with AWS but also between VTS and Bluecurrent,” says Patrick Cheffins, head of product for Diverge at VTS. “It wouldn’t have been possible without all of us working together.”

In order to ingest and store meter data at scale, Diverge uses services such as [AWS Lambda](#), a serverless compute service, and [Amazon DynamoDB](#), a serverless, NoSQL, fully managed database. It also uses new, purpose-built AWS services, including an energy data storage solution developed as part of the co-development effort. Diverge is processing six times the data ten times faster than Bluecurrent’s previous solution. It also has advanced features supporting efficient operations in Australia’s complex metering market, including an improved user interface and advanced tools for individual and bulk exception resolution.

The Diverge solution help Bluecurrent store the data that it ingests on a new service developed by AWS as part of the strategic alliance specifically for storing energy data at scale and with high performance. Diverge then securely delivers the designated data to each of Bluecurrent’s customers, including to local distribution networks in order to provide visibility of power quality at the edge of the grid, to energy retailers for billing and enhanced customer service, and to electricity market operators for market settlement. Using Diverge, Bluecurrent is meeting stringent new Australian regulatory requirements and commitments to individual customers, delivering meter data hours earlier and more reliably. “Now on AWS, Bluecurrent is delivering six times the data ten times faster,” says Dean Van Gerrevink, general manager of Diverge for VTS.

Bluecurrent is also more resilient because of the scalability of Diverge. Performance improvements mean the company can recover from any outage faster. “Because Diverge is cloud native and serverless, we can automatically scale to meet any increase in demand and then resume normal service automatically,” says Cheffins.

Outcome | Adapting to Meet Evolving Energy Industry Challenges, Now and in the Future

Using Diverge, Bluecurrent is equipped to adjust to future changes in the energy industry and to support customers with new capabilities. “The teams from VTS, AWS, and Bluecurrent worked together to achieve what the business needed to achieve—not only delivering a new system but working to enhance it and resolve the exceptions that come up in normal day-to-day work,” says Carrick. “We have flexibility in terms of where we’re going and the breadth of services and solutions Bluecurrent can now offer to its customers.”

By using a microservices architecture on AWS, VTS can adapt the solution to future changes in the energy industry increasingly relies on data to manage the effects of electrification and renewable generation. For Bluecurrent, this meant delivering a lot more data much faster to more recipients. “Going from meter consumption in 30-minute increments to 5-minute in Australia alongside a range of other new data and event



information with varying frequencies required a higher performance solution. Diverge has already demonstrated a 200-times increase in raw throughput,” says Cheffins. “It’s important to have services such as Diverge that are fit for this purpose and that meet both the functional and regulatory requirements of these use cases now and into the future.”

“Electrification and the supply intermittency of renewables are real challenges in the sector,” says Cheffins. “We’re passionate about building this energy data solution to meet these challenges for our customers and society.”

About Bluecurrent

Bluecurrent is a smart meter services company providing data and innovative services to energy retailers and distributors since 2007. Bluecurrent has a fleet of over two million meters across Australia and New Zealand.

AWS Services Used

Amazon DynamoDB

Amazon DynamoDB is a serverless, NoSQL database service that enables you to develop modern applications at any scale.

[Learn more »](#)

AWS Lambda

AWS Lambda is a compute service that runs your code in response to events and automatically manages the compute resources, making it the fastest way to turn an idea into a modern, production, serverless applications.

[Learn more »](#)

More Energy Customer Stories

EUROPE, MIDDLE EAST, & AFRICA

UNITED STATES





Jimmy Energy Builds Scalable and Secure HPC...

Jimmy Energy (Jimmy) has been turning up the heat, with nuclear power taking center stage in its massive decarbonization campaign. Its small-scale generators can reduce carbon footprints across multiple industries, but to facilitate their safe and efficient operation, the company needed a reliable high performance computing (HPC) solution. Using Amazon Web



2024



Conduit Power Slashed SCADA Deployment Time...

Learn how independent power producer Conduit Power implemented a resilient cloud-based SCADA system using AWS.

2024

SPAIN



Optimizing SAP Performance and...

Learn how integrated energy company Repsol optimized its SAP systems using AWS.

2024

SPAIN



Iberdrola Supports Customers' Energy...

Using AWS services, global energy company Iberdrola built the Advanced Smart Assistant to help customers reduce smart device energy consumption and better manage grid consumption and capacity.

2023



Get Started

Organizations of all sizes across all industries are transforming their businesses and delivering on their missions every day using AWS. Contact our experts and start your own AWS journey today.

Contact Sales

Learn About AWS

- What Is AWS?
- What Is Cloud Computing?
- AWS Accessibility
- AWS Inclusion, Diversity & Equity
- What Is DevOps?
- What Is a Container?
- What Is a Data Lake?
- What is Artificial Intelligence (AI)?
- What is Generative AI?
- What is Machine Learning (ML)?
- AWS Cloud Security
- What's New
- Blogs
- Press Releases

Resources for AWS

- Getting Started
- Training and Certification
- AWS Solutions Library
- Architecture Center
- Product and Technical FAQs
- Analyst Reports
- AWS Partners

Developers on AWS

- Developer Center
- SDKs & Tools
- .NET on AWS
- Python on AWS
- Java on AWS
- PHP on AWS
- JavaScript on AWS

Help

- Contact Us
- Get Expert Help
- File a Support Ticket
- AWS re:Post
- Knowledge Center
- AWS Support Overview
- Legal
- AWS Careers



Language

- عربي |
- Bahasa Indonesia |
- Deutsch |
- English |
- Español |
- Français |
- Italiano |
- Português |
- Tiếng Việt |
- Türkçe |
- Русский |
- ไทย |
- 日本語 |
- 한국어 |
- 中文 (简体) |
- 中文 (繁體)

Privacy

- |
- Accessibility

- |
- Site Terms

- |
- Cookie Preferences

|

© 2024, Amazon Web Services, Inc. or its affiliates. All rights reserved.

